Sensors Are Messengers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Sensors Are Messengers

> Prepare Sensors Are Messengers: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Describe a **measurement** using its quantity, value, **unit** and conditions

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Input, Process, Output

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

Look for: Trace information through a system without confusing information with a physical action

Reference example, not the only acceptable wording: **Input** includes both request fields and all three catalogue records. The **process** checks science AND pages<=120.

Output contains A only; B fails the length condition and C fails the genre condition. Listing A does not establish that a copy was reserved.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

Concept to explain

Sensors measure **signal**s; later **process**ing gives those **signal**s meaning.

Term	Meaning
sensor	device that measures a physical signal
signal	a changing quantity that can be measured
measurement	a recorded value describing part of a signal
data	represented information available for processing

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal **data** and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Separate physical quantity, recorded **measurement** and interpretation throughout. No hardware is required. Avoid describing every difference as error or every repeated value as accuracy.

Consistent disagreement is a finding; deciding which device is accurate requires additional reference **evidence**. Keep **unit**s visible in both speech and written work.

Readiness answer

No. A maximum of 26 degrees C could occur at noon or at 3 pm. The retained number does not preserve time.

The paired-temperature table and two paper device labels

A sketch of a classroom with a window, shaded wall, speaker and door

Optional ordinary room thermometers used according to school procedures; paper **data** is sufficient

End goal for the pictorial case

By the end, I can compare **sensor** readings while controlling where and when they are measured.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Sensors Are Messengers using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT light, sound, or heat from the environment

2. Process — the deciding step

PROCESS a **sensor** converts the **signal** into recorded values

3. Output — the result

OUTPUT numbers or pixel/sound **data** sent to a later **process**

Key idea

CORE IDEA **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.

How the mechanism works

UNDER THE HOOD A **sensor** is a transducer: it converts a physical quantity into a **signal** a computing system can represent. Sensing and interpretation are separate. A camera measures light; an image model may later infer a label.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A classroom weather station measures light, sound, and temperature but does not decide what those **measurement**s mean.

Step 1

STEP 1 — OBSERVE light, sound, or heat from the environment

Step 2

STEP 2 — TRACE a **sensor** converts the **signal** into recorded values

Step 3

STEP 3 — CONCLUDE numbers or pixel/sound **data** sent to a later **process**

Model answer

MODEL REASONING 820 could be a light **measurement**, 24 C comes from a temperature **sensor**, and changing sound values come from a microphone.

The numbers alone do not prove whether the plant needs water or whether a voice was present; later **process**ing must interpret them.

A claim to avoid

DO NOT OVERCLAIM Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

Which measures a light-related **signal**? Can either directly measure happiness?

Exact explanation to model

Why the result follows

The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

Explain the added worked example

Explain the added worked example · 1

A garden display shows 24. Is that warm? We cannot interpret the number until we know what was measured and the **unit**.

It might mean 24 degrees Celsius, 24 percent soil moisture according to a particular device, or 24 hours since watering. The screen alone does not supply the missing meaning.

Explain the added worked example · 2

A useful **measurement** record names the quantity, value, **unit**, time and relevant conditions. For air temperature, the location and exposure matter: a device in direct sunshine may warm differently from one in shade.

Record the conditions rather than assuming that every nearby device measured the same situation.

Measurement is a way of representing part of the physical world. It does not capture everything about that world.

One air-temperature reading does not tell us how dry the roots are, whether a plant is diseased or whether watering is permitted.

The **sensor** responds to a physical quantity. Electronics convert the response into a recorded value. A later program can compare the value with a threshold.

Nothing about this path requires us to claim that the **sensor** understands plant health.

The rule uses exceeds 28, so a reading of exactly 28 does not trigger it. At 29 it does.

This boundary is a property of the **written rule**; it is separate from the question of whether the measuring device is accurate enough for the intended purpose.

Key terms

Term	Meaning
Sensor	A device that responds to a physical quantity and helps turn it into a signal
Measurement	A value obtained by comparing a quantity with a scale or standard
Unit	An agreed amount used to express a quantity, such as a second or a degree Celsius

Term	Meaning
Variation	Differences among readings or observations
Calibration	Checking and, when appropriate, adjusting a measuring system against a reference

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 3

Two devices are placed side by side in a shaded room and allowed to settle. At three agreed times, device A reads 20, 21 and 20 degrees C; device B reads 22, 23 and 22 degrees C. B is 2 degrees higher on every comparison.

That pattern is **evidence** of a consistent difference under these test conditions.

Time	A in degrees C	B in degrees C	B minus A
10:00	20	22	2
10:05	21	23	2
10:10	20	22	2

Model the reasoning aloud · 4

We cannot conclude from agreement patterns alone that A is correct and B is wrong. A suitable checked reference can help investigate.

We also need to consider whether the devices have different response times, resolutions or placement. Three comparisons in one room do not prove performance in every temperature range.

Calibration is a reference check, sometimes followed by an adjustment. Do not simply subtract 2 from every future B reading because these three pairs happened to differ by 2.

A correction needs support over the conditions where it will be used.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sensors Are Messengers

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

The facts we will use

Supplied facts

On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Supplied facts

Neither device has been compared with a trusted reference.

AI ACADEMY / AI-1/3

Simultaneous light readings

	Sensor A	Sensor B	B minus A
Time 1	40	45	5
Time 2	50	55	5
Time 3	60	65	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/3

Sensors Are Messengers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Same shelf

Place both devices together and allow them to settle.

2. Paired readings

Record three matching times: (40,45), (50,55), (60,65).

3. Compare gaps

B minus A is 5, 5 and 5 units.

4. Check a reference

A trusted reference is needed to decide which reading is closer.

CHANGE ONE THING

Move only B into brighter light.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Same shelf	Place both devices together and allow them to settle.
2. Paired readings	Record three matching times: (40,45), (50,55), (60,65).
3. Compare gaps	B minus A is 5, 5 and 5 units.
4. Check a reference	A trusted reference is needed to decide which reading is closer.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently.

What this example does not establish

Keep the stated limit

Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

Change one thing in the pictured case

Change one thing

Change: Move only B into brighter light.

Predicted result

Prediction: The next gap could change for reasons other than a difference in device accuracy.

Why the result follows

Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Sensors Are Messengers, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

2. State the decision to explain

Write the question to explain: Which measures a light-related **signal**? Can either directly measure happiness?

3. Reveal the worked reasoning

Reveal after **prediction**s: The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

Hold back the independent task

Keep for an independent attempt: A station reports light level 18 and temperature 22 degrees C. Which **sensor** produced each reading? Can either reading alone tell whether a child feels comfortable? Explain.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Put the devices together, use comparable settings and times, allow settling and repeat, or swap positions while recording conditions.

If the difference follows position, location is plausible; if it follows device, investigate the device or settings.

For the exit: ask what quantity and **unit** 30 refers to, when and where it was measured, and whether that **measurement** is suitable for the rule.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve information loss with the maximum-temperature prompt. Ask what time information disappeared.

Stage	Teacher action
7 to 20 minutes	Show only 24. Collect possible meanings, then build a measurement record with unknown fields.
20 to 33 minutes	Trace the garden diagram and test 27, 28 and 29. Separate sensing from the rule.
33 to 45 minutes	Pairs complete W1 and W2. Reject invented record details while accepting explicit unknowns.
Session 2 start 0 to 8 minutes	Compute paired differences in the worked table. Ask whether consistent disagreement identifies the correct device.
8 to 23 minutes	Model a location comparison. Complete W3 and discuss what a reference adds.
23 to 38 minutes	Collect W4 independently. Ask for two predicted outcomes, not just a procedure.
38 to 45 minutes	Use W5 and the exit question to distinguish measurement from an unsupported plant-health claim.

Reduce help deliberately

Completion practice

Completion task: A warning rule is recorded temperature greater than 28 degrees C. State its **output** for 27, 28 and 29 degrees C. Explain why an off warning does not prove that the **sensor** is working correctly.

Reasoning to listen for

Teacher check: 27 off, 28 off, 29 on. These **output**s follow the comparison applied to the recorded values.

A device could report an incorrect low value while the rule correctly stays off; **measurement** quality needs a separate check.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Sensor**s and **measurement**. Proposed explanation: “The more stable device must be accurate”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: A device can repeatedly report an offset value. Stability and closeness to a reference answer different questions.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding **variation**.

Pictorial practice question

> Why can we calculate the earlier gap but not predict an exact new B reading?

Reasoned teacher answer

The earlier pairs give 5 each time. No **measurement** or light-response rule is given for B’s new location.

Changed-case reasoning: The next gap could change for reasons other than a difference in device accuracy. The comparison now changes location as well as device; there is no supplied reading from the new location.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Sensors Are Messengers investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Match camera, microphone, and temperature **sensor**s to **data**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR A **sensor** knows what a person feels.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The more stable device must be accurate: A device can repeatedly report an offset value. Stability and closeness to a reference answer different questions.

The warning proves the condition in the world: Trace the recorded number into the rule. An incorrect or irrelevant **input** can produce a correctly computed warning.

Missing details can be filled with sensible guesses: Keep unknown fields explicit until **evidence** supplies them. Guesses must be labelled as assumptions.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Sensors Are Messengers questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

Independent question

Independent question · withhold the key

A station reports light level 18 and temperature 22 degrees C. Which **sensor** produced each reading? Can either reading alone tell whether a child feels comfortable? Explain.

Expected independent answer

Teacher key · Expected independent answer

A light **sensor** produced 18; a temperature **sensor** produced 22 degrees C. Neither directly measures a child's comfort. Ask the person rather than treating a **measurement** as their feelings.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **sensor** knows what a person feels.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

No: scales, **calibration** and settings may differ. Use the same calibrated **sensor** and settings in both rooms, record **unit**s and repeat at comparable positions and times.

A larger uncalibrated number alone cannot establish a brightness ratio.

Explanations for the additional workbook examples

W1 Repair an incomplete record

Workbook question · W1

A notebook entry says 24 at lunchtime. Rewrite it as a useful air-temperature record by stating which missing fields must be collected. Do not invent values that were not observed.

Teacher reasoning · W1

Keep the observed value 24 and approximate time lunchtime; mark the **unit**, exact time if needed, location, exposure and device identity as unknown or to be checked.

A suitable completed record needs air temperature, value, **unit** and relevant conditions. Invented degrees C or shade is not an **observation**.

W2 Trace a boundary

Workbook question · W2

A warning rule is recorded temperature greater than 28 degrees C. State its **output** for 27, 28 and 29 degrees C. Explain why an off warning does not prove that the **sensor** is working correctly.

Teacher reasoning · W2

27 off, 28 off, 29 on. These **output**s follow the comparison applied to the recorded values. A device could report an incorrect low value while the rule correctly stays off; **measurement** quality needs a separate check.

W3 Compare without choosing a winner

Workbook question · W3

At the same three times, A records 18, 19 and 18 degrees C and B records 21, 22 and 21 degrees C in the same location. Calculate the paired differences. What is established, and what further **evidence** would help judge accuracy?

Teacher reasoning · W3

B minus A is 3, 3 and 3 degrees C. This establishes a consistent observed difference in these comparisons. It does not show which device is closer to the actual temperature.

Use an appropriate checked reference, allow settling, record conditions and investigate across the intended range.

W4 Investigate a new measurement dispute

Workbook question · W4

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

Teacher reasoning · W4

Place both devices side by side under the same normal sound conditions, using the same settings and reading times, or exchange positions and repeat.

If readings become similar together or the higher reading follows the speaker location, location is a plausible explanation.

If one device remains consistently lower side by side or after swapping, a device or setting difference remains plausible. Neither result alone proves a device broken; a checked reference and repeated **observation**s help.

W5 Limit a system's conclusion

Workbook question · W5

An air-temperature **sensor** reports 29 degrees C and a rule turns on a warning. Someone writes The plant is thirsty.

Explain the unsupported step and name information more directly relevant to deciding whether watering is needed.

Teacher reasoning · W5

Crossing an air-temperature threshold does not establish root-zone dryness or a plant's watering need. Soil conditions, the kind of plant, recent watering and a gardener's inspection are more relevant.

The warning can be correctly on while the thirst claim remains unsupported.

Assess independent transfer

Independent question · withhold the key

Selected case: A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

Selected case answer

Teacher key · Selected case answer

Place both devices side by side under the same normal sound conditions, using the same settings and reading times, or exchange positions and repeat.

If readings become similar together or the higher reading follows the speaker location, location is a plausible explanation.

If one device remains consistently lower side by side or after swapping, a device or setting difference remains plausible. Neither result alone proves a device broken; a checked reference and repeated **observation**s help.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Sensors Are Messengers to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 4: Rules or Learned Patterns?. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Give a record frame with quantity, value, **unit**, time and location boxes. Use a number line for paired differences.

Extension: discuss why devices that round to whole degrees can differ at a boundary without either being grossly faulty.

W4 is secure when it controls location or device identity in a meaningful comparison, keeps ordinary safe conditions, states two interpretable outcomes and avoids declaring certainty.

If the learner only says try again, ask which factor will change and what each possible result would mean. Retest independently with light readings at a desk and window.

Week 4 uses reliable recorded features to compare an explicitly **written rule** with a rule selected from labelled examples.

Plan the delayed return

Suggested return points in this level: ai-1/4, ai-1/6, ai-1/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Sensors Are Messengers lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

> Teaching target: Describe a **measurement** using its quantity, value, **unit** and conditions

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A warning rule is recorded temperature greater than 28 degrees C. State its **output** for 27, 28 and 29 degrees C.

Explain why an off warning does not prove that the **sensor** is working correctly.

Supported practice answer

Reasoned practice answer: 27 off, 28 off, 29 on. These **output**s follow the comparison applied to the recorded values.

A device could report an incorrect low value while the rule correctly stays off; **measurement** quality needs a separate check.

Check every learner before discussion: Use the worked case for **Sensor**s and **measurement**. Proposed explanation: “The more stable device must be accurate”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A device can repeatedly report an offset value. Stability and closeness to a reference answer different questions.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Move only B into brighter light. Require a **prediction** and a reason before comparing. Expected result: The next gap could change for reasons other than a difference in device accuracy.

Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

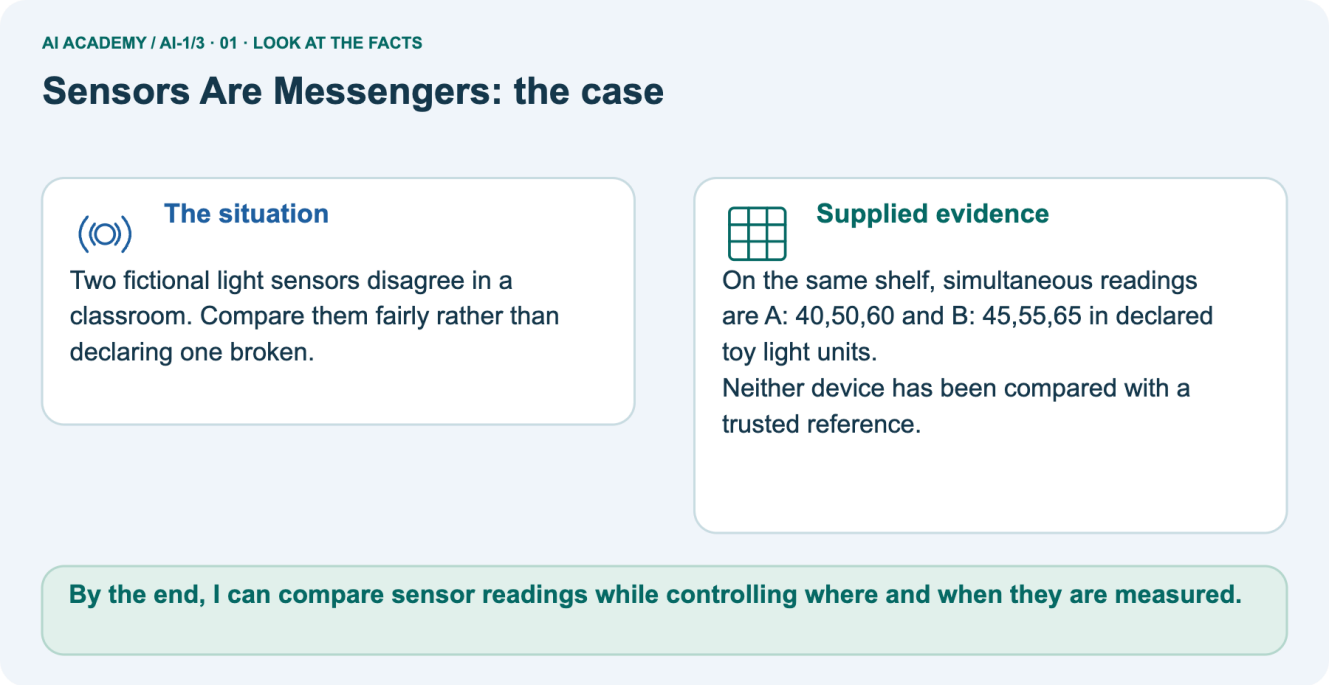
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Sensors Are Messengers graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sensors Are Messengers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken. Supplied facts: On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

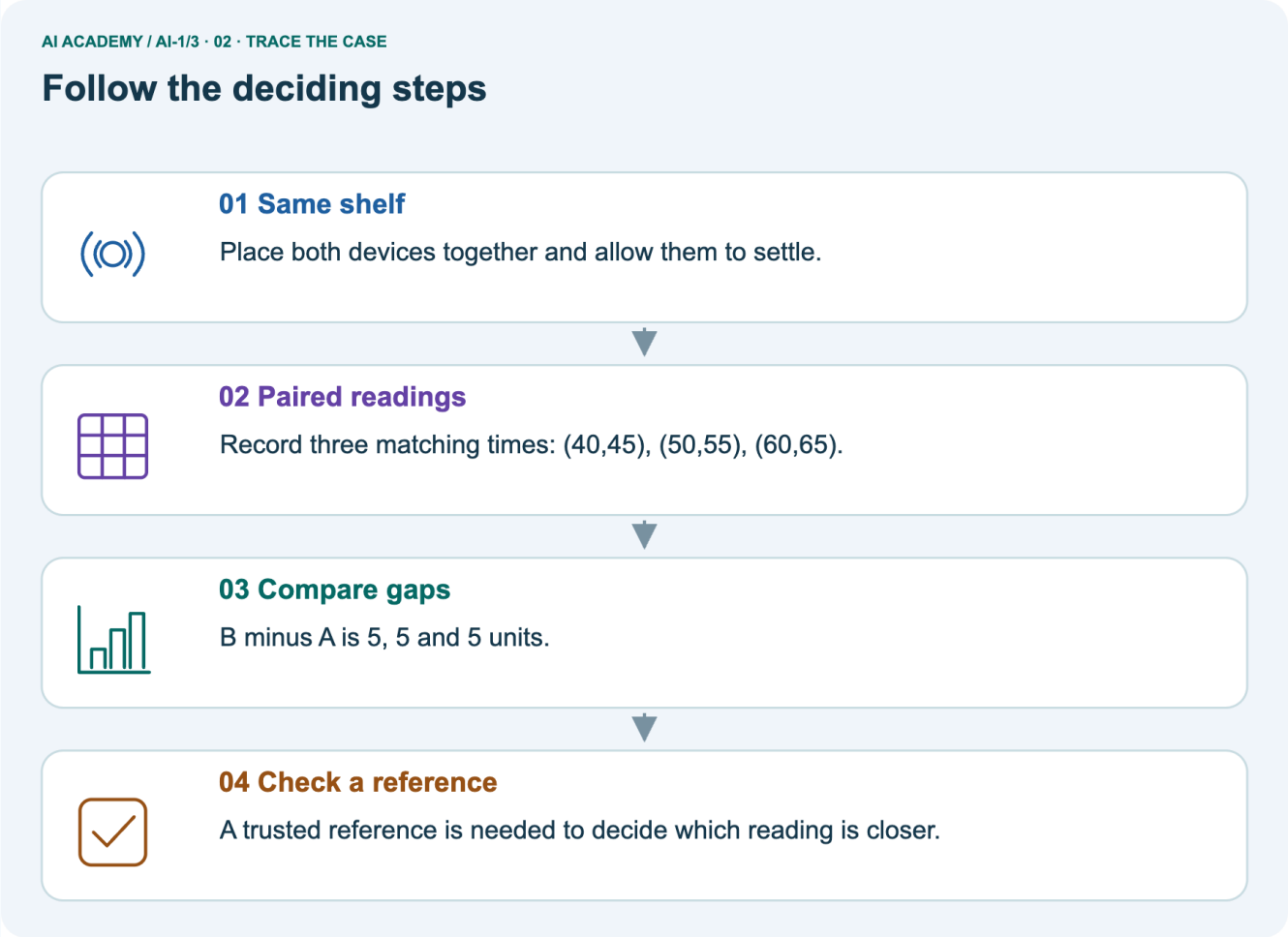
Neither device has been compared with a trusted reference.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

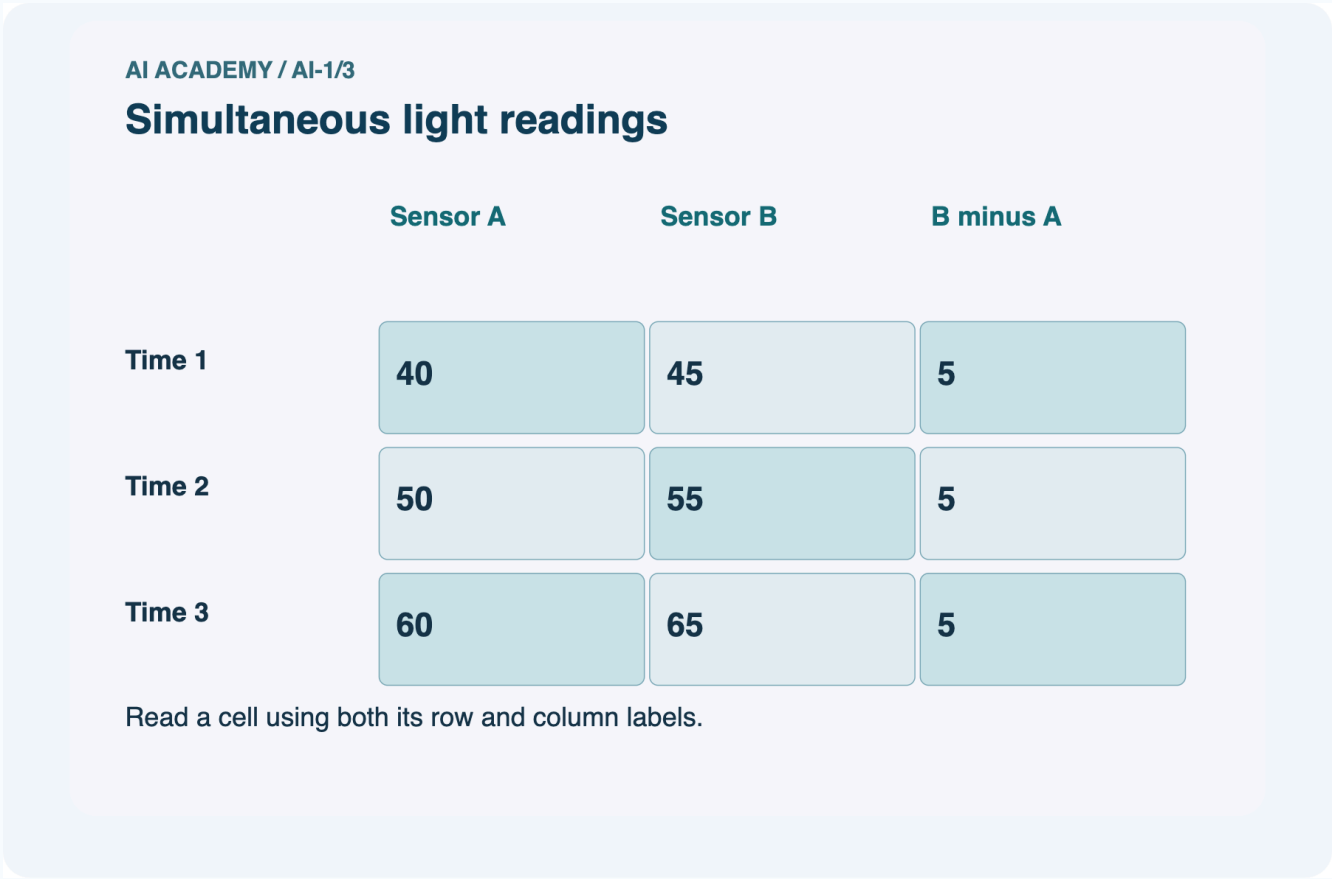
Same shelf: Place both devices together and allow them to settle. → Paired readings: Record three matching times: (40,45), (50,55), (60,65). → Compare gaps: B minus A is 5, 5 and 5 **unit**s. → Check a reference: A trusted reference is needed to decide which reading is closer.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Simultaneous light readings



Follow the labelled picture with the matching explanation. Open the vector diagram

Simultaneous light readings. Time 1: **Sensor** A = 40, **Sensor** B = 45, B minus A = 5. Time 2: **Sensor** A = 50, **Sensor** B = 55, B minus A = 5. Time 3: **Sensor** A = 60, **Sensor** B = 65, B minus A = 5.

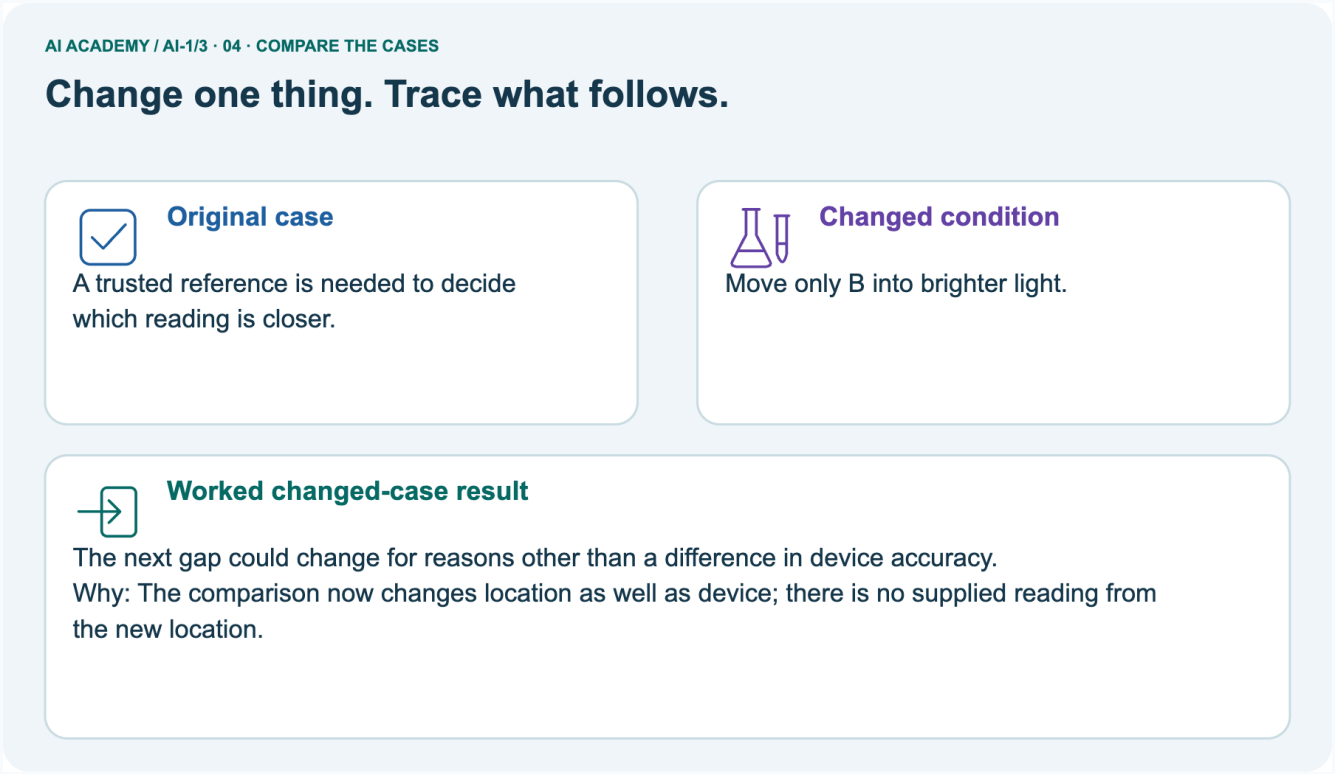
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, **unit**s and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A trusted reference is needed to decide which reading is closer. Changed condition: Move only B into brighter light.

Expected result: The next gap could change for reasons other than a difference in device accuracy. Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

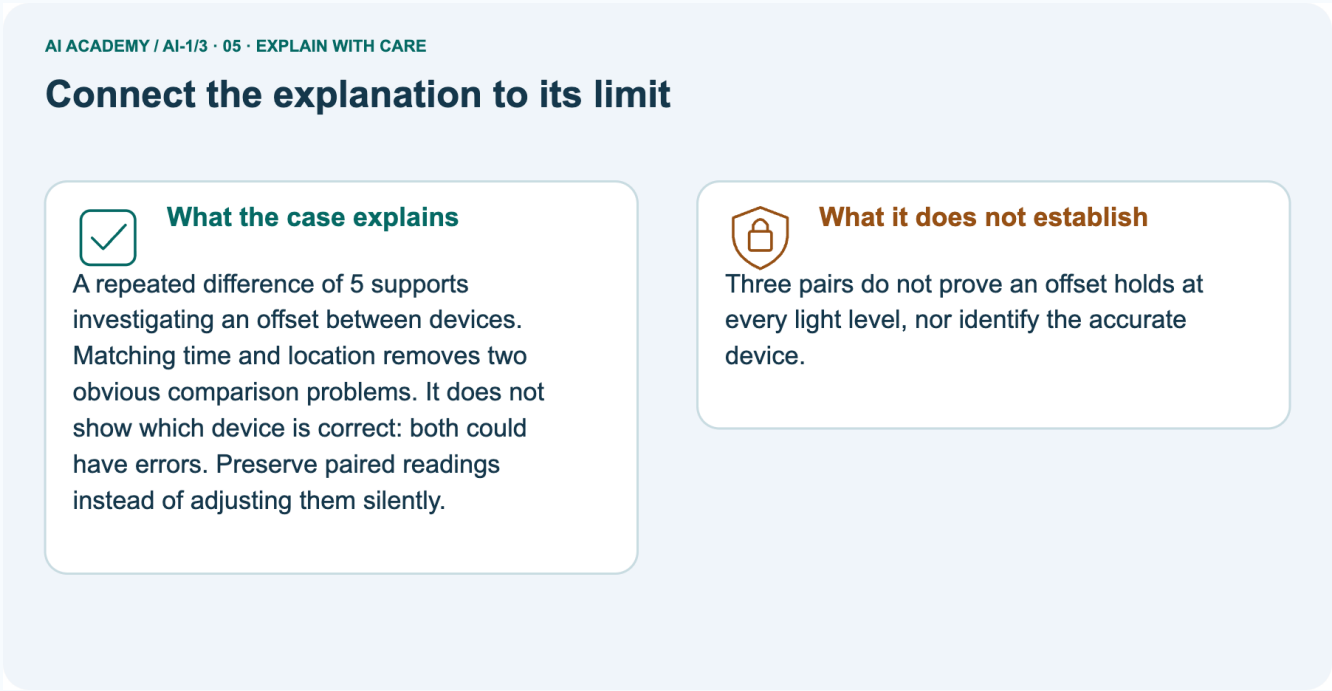
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The comparison now changes location as well as device; there is no supplied reading from the new location.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently. Limit: Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

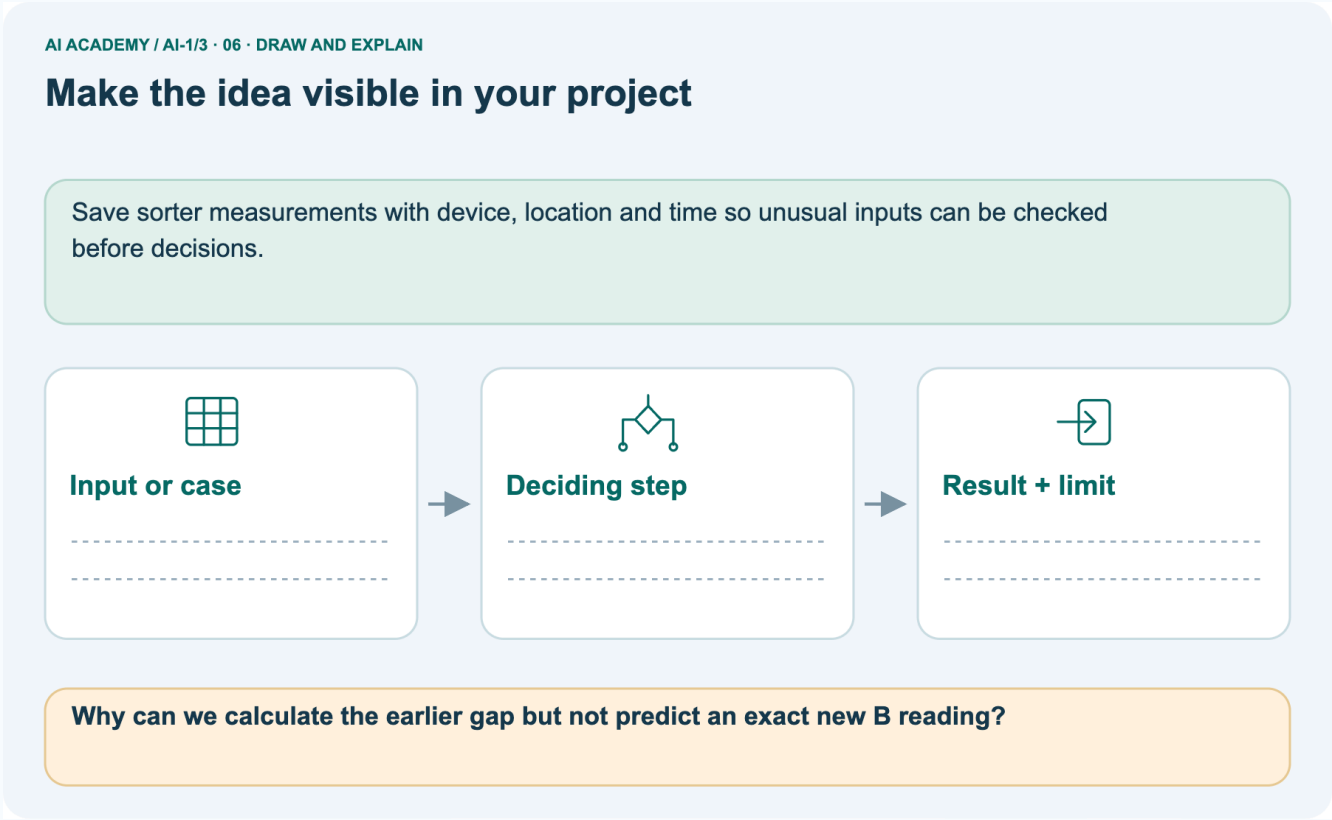
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions. Draw your **input**, deciding step, result and limit.

Supported question: Why can we calculate the earlier gap but not predict an exact new B reading?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The earlier pairs give 5 each time. No **measurement** or light-response rule is given for B′s new location.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.